Do Now · retrieval

Denary → floating point

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Choose a type

Q2

Convert a hexadecimal byte

Q3

Use two’s complement

Q4

Explain a fixed-width shift

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Binary fractions continue right of the point — CODE:
6.5 = 4 + 2 + ½
= 110.1_2

0.375 = ¼ + ⅛
= 0.011_2

Core idea 2

Floating point uses two signed fields — The mantissa stores the signed significant fraction. The exponent stores the signed power-of-two scale.

Core idea 3

Normalise the mantissa — For a non-zero value, a positive normalised mantissa begins 01 ; a negative one begins 10 . — If redundant 00 or 11 sign bits remain, shift the mantissa left and decrease the exponent by the same number of places.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Precision and range — More mantissa bits improve precision . More exponent bits increase range .

Vocabulary · use the correct technical terms

Mantissa — The signed fractional field holding the significant binary digits.

Exponent — The signed integer that specifies the power-of-two scale.

Binary point — The boundary between integer and fractional places.

Normalisation — Removing redundant leading sign bits while preserving the represented value.

Independent skill check · apply the lesson

Represent -0.375 in normalised floating-point form using an 8-bit two's-complement mantissa and 4-bit exponent. Show each step.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Represent +6.5 in normalised floating-point form

Follow the worked method

1. Write 6.5 in binary — CODE:
$6.5 = 110.1_2$ — The whole-number part is 110 ; the 0.5 fractional part is .1 .

2. Move the binary point — CODE:
110.1_2
$= 0.1101_2 \times 2^3$ — The binary point moves three places left, so the exponent is 3 .

3. Write the two fields — CODE:
Mantissa: 01101000
Exponent: 0011

Stored: 01101000 0011 — Check: $0.8125 \times 8 = 6.5$.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: 0.15625 is 1/8 + 1/32. Write this as a binary fraction.

Clue 2: For a normalised positive two's-complement mantissa, the first two bits are 01.

Clue 3: If the binary point moves left, compensate using a negative exponent.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Convert the fraction to binary.

Step 2: Normalise the fraction.

Step 3: Encode the 8-bit mantissa and signed 4-bit exponent.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Represent positive values

Represent (a) 10.5 and (b) 0.375 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show the binary conversion and both fields. [2 each]

Adapted from the OCR 2023 conversion task: values changed; classroom marks.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Handle a boundary value

Represent −0.5 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the mantissa [1] and exponent [1].

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

More mantissa bits improve floating-point precision.

Use key vocabulary from memory

1. Explain Mantissa in your own words.

2. Explain Exponent in your own words.

3. Explain Binary point in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

0.15625 = 0.00101 (binary) = 0.101 x 2^-2; mantissa 01010000, exponent 1110.

PAGE 3 · Independent skill check:

-0.375 = -0.11 binary = -0.75 x 2^-1. Mantissa 10100000; exponent 1111. Normalised sign bits are 10.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-004/answers.html